Circuit Simulation Lab:19
Design of Counters

The purpose of this experiment is to introduce the design of Synchronous sequential circuits, in this case we are taking an 8 bit down counter.

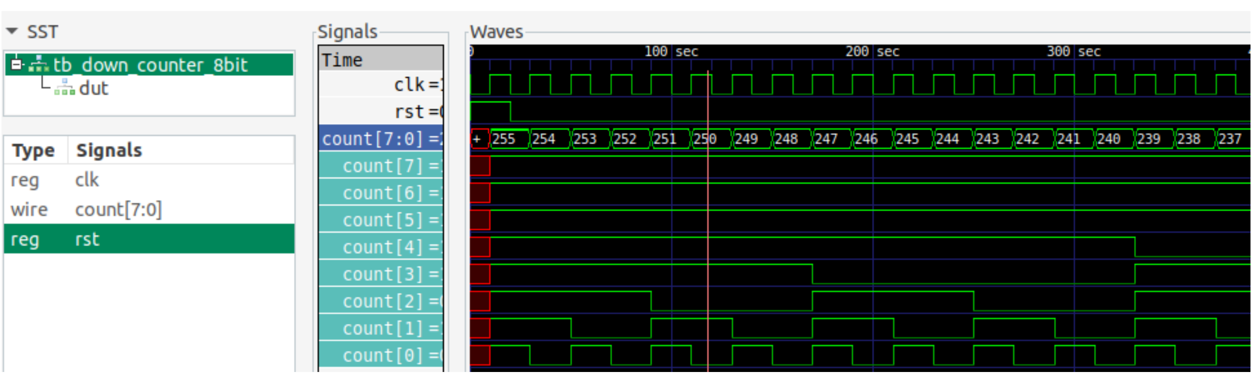
CODE:

```
module down_counter_8bit(
input clk,
input rst,
output reg [7:0] count
);

always @(posedge clk)
begin

if(rst)
   count <= 8'd255;

else
   count <= count - 1'b1;

end

Endmodule
```

TESTBENCH

```
module tb_down_counter_8bit;

reg clk;
reg rst;

wire [7:0] count;

down_counter_8bit dut(
.clk(clk),
.rst(rst),
.count(count)
);
```

```
always #10 clk = ~clk;

initial begin

$dumpfile("down_counter.vcd");
$dumpvars(0,tb_down_counter_8bit);

clk = 0;
rst = 1;

#20;
rst = 0;

#500;

$finish;

end

initial begin
$monitor("time=%0t rst=%b count=%d",
        $time,rst,count);
end

Endmodule
```



```
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_19$ vvp out
VCD info: dumpfile down_counter.vcd opened for output.
time=0 rst=1 count=  x
time=10 rst=1 count=255
time=20 rst=0 count=255
time=30 rst=0 count=254
time=50 rst=0 count=253
time=70 rst=0 count=252
time=90 rst=0 count=251
time=110 rst=0 count=250
time=130 rst=0 count=249
time=150 rst=0 count=248
time=170 rst=0 count=247
time=190 rst=0 count=246
time=210 rst=0 count=245
time=230 rst=0 count=244
```